Distributed Systems

3rd Lecture

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Outline

- Clock and Global States
  - Global States
  - Determining Consistent Global States
- Q&A

Global States

- Prefix of Pi's History & Global History

$$h_i^k = < e_i^j \mid j = 1, ..., k >, \quad H = \bigcup_{i=1}^{N} h_i$$

- Cut & Frontier

$$C = \bigcup_{i=1}^{N} h_i^{C_i}, \quad F = \{e_i^{C_i} \mid i = 1, ..., N\}$$

Set of All Affected Values

- Global State (Corresponding to C)

$$S = \{s_i^{C_i} \mid i = 1, ..., N, s_i^{C_i} \text{ is } P_i\text{'s state immediately after } e_i^{C_i}\}$$

- Run: a Total Ordering in a Global History Consistent with Each Local History

Consistent Cuts & Runs

- C Is Consistent If the Following Holds:

$$\forall e \in C,\ e' \rightarrow e \ \Rightarrow\ e' \in C$$

H Is Consistent If the Corresponding C Is Consistent

Figure 14.9 on Page 629

- Consistent Run: a Total Ordering in a Consistent Global History, Consistent with the Happened-Before Relation

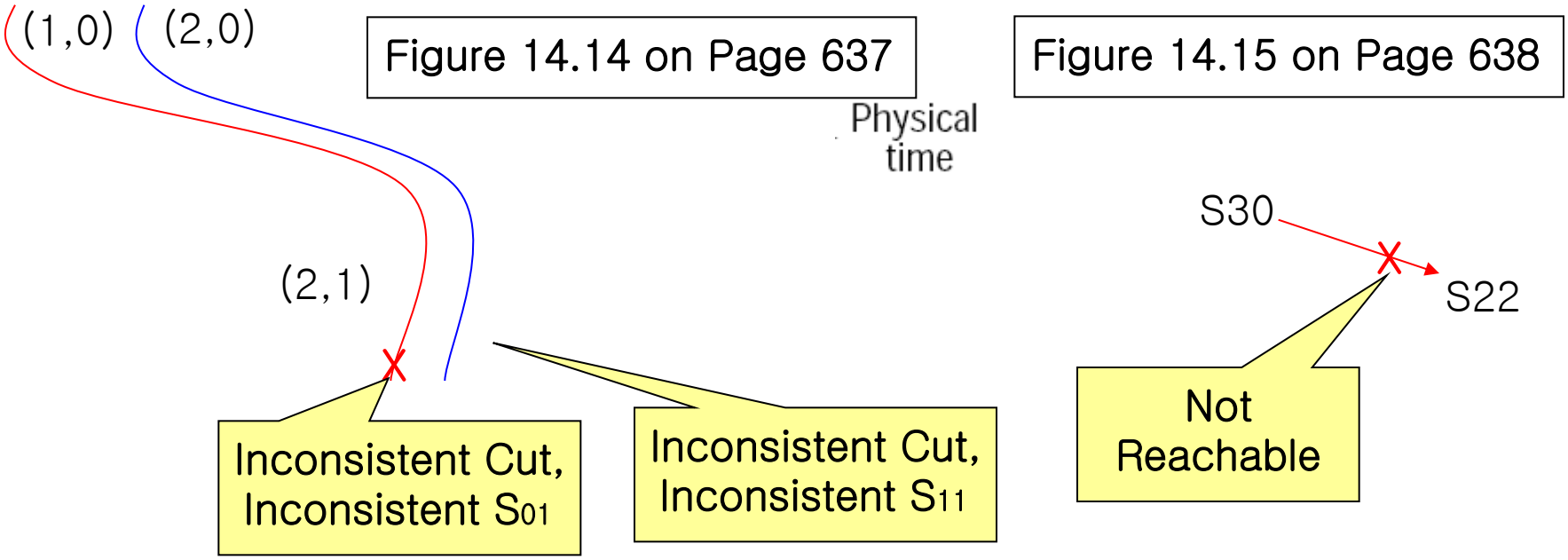
Lattice of Global States

Observing Consistent Global States

$$S = \{s_i \mid i = 1, ..., N\} \textit{ Is Consistent iff } VC_i(s_i)[i] \geq VC(s_j)[i] \textit{ for } i, j = 1, ..., N$$

Sij Is in Level (i+j)

S_{ij}= global state after *i* events at process 1 and *j* events at process 2



Detecting Global Properties

Figure 14.8 on Page 627

Distributed 'Snapshot' Algorithm [Chandy85]

- Consistent Global-State Detection

Marker to Record the State

Assumptions
- Reliable, Strongly-Connected Components
- Unidirectional Channels & In-Order Message Delivery

Figure 14.10 on Page 632

Illustration: How the Alg. Works

- Initial States of the Components



Figure 14.11 on Page 632

P2 Has Already Received an Order of Five Widgets

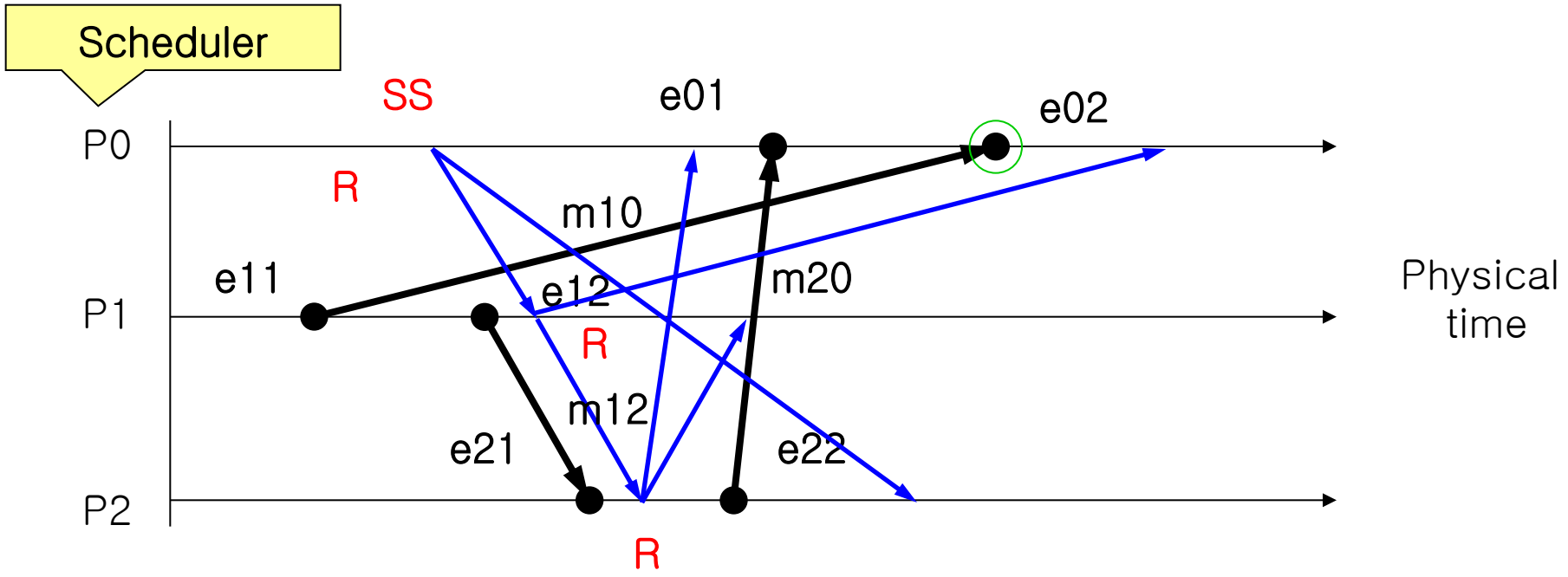
Illustration (Cont'd)

P2 Has Already Received an Order of Five Widgets

Figure 14.12 on Page 633

- S = p1: <$1000,0>; p2:<$50,1995>;
  c1: <(five widgets)>; c2:<>

Illustration w/ a Diagram



- SS: p0: <>; p1: <e11,e12>; p2:<e21>
  c01:<>; c02 <>; c10<m10>; c12 <>
  c20 <>; c21<>

Consistency Proof

- States Recorded by the Alg. Are Consistent:

$$\forall e_j \in C,\ e_i \rightarrow e_j \ \Rightarrow\ e_i \in C$$

$$Show:\ e_i \notin C,\ e_i \rightarrow e_j \ \Rightarrow\ e_j \notin C$$

i≠j

  - Assume That P_i Recorded Its State before e_i
  - Marker Would Have Reached P_j before the Message for e_j
  - P_j Would Have Recorded Its State before e_j

Characteristics of Snapshots

- Derivation of "Observed" Run from "Actual" Run

$$"Actual" \ Run : < e_i^k \mid i = 1, ..., N > = < e^j >$$

$$"Permuted" \ Run : < ..., e^{R-1}, e^R, ... >$$

Recodring

Observed (Consistent) Run

  - A Non-Observed Event May Occur before an Observed Event in the "Actual" Run
  - If a Non-Observed Event Precedes an Observed Event (Next to it) in the "Actual" Run, the Events Can Be Swapped Preserving Consistency

Global State Predicates

- Functions That Map Global States to True or False
  - Stable: Once True, It Remains True
    - E.g., deadlock or termination
  - Unstable: Not Stable
    - Possibly True: True At Some Point
      - E.g., snapshot by the 'Snapshot' Algorithm
    - Definitely True: True in All Cases